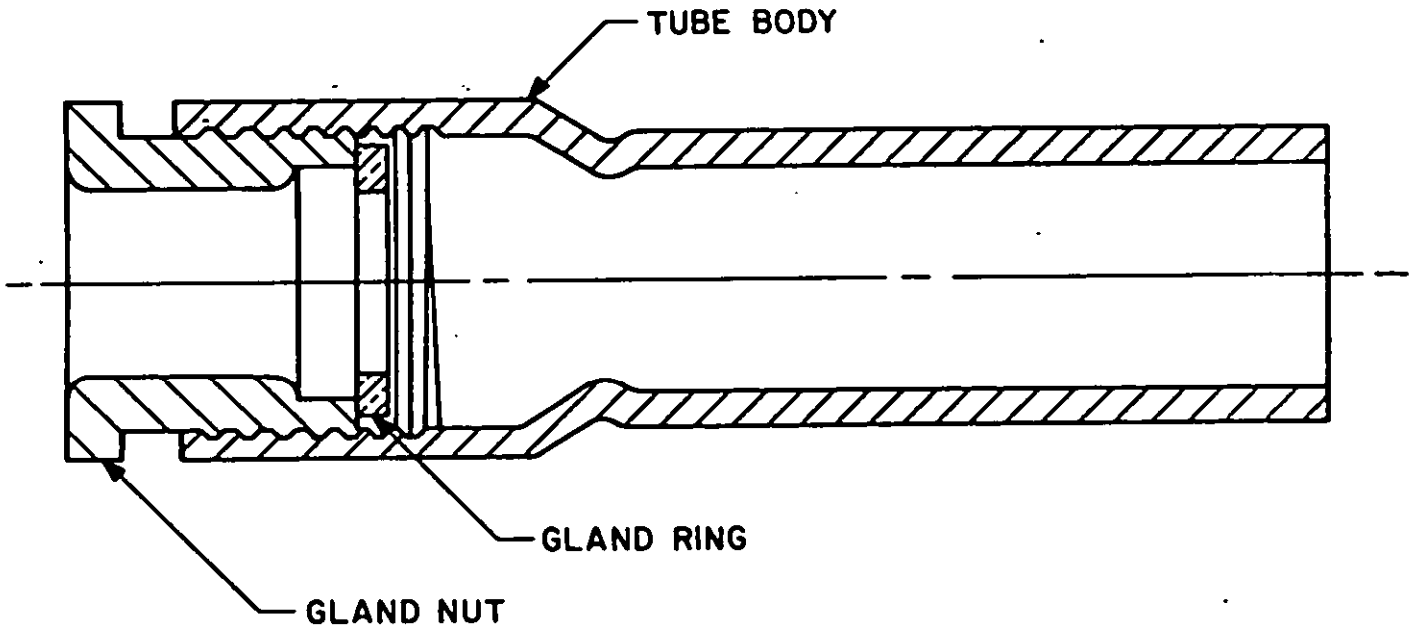
INCH-POUND
MIL-S-24235/18(SH)
21 August 1989

MILITARY SPECIFICATION SHEET

# STUFFING TUBES, METAL, AND PACKING ASSEMBLIES FOR ELECTRIC CABLES, SWAGE TYPE, REDUCED DIAMETER, STEEL AND ALUMINUM, FOR DECK AND BULKHEADS WITH PIPE PROTECTION

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24235.




SH 131865

FIGURE 1. Stuffing tube assembly. (01)

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances shall be as follows:
   Decimals - plus or minus 0.005 inch.
   Fractions - plus or minus 1/64 inch.
   Angular dimensions - plus or minus 30 minutes.

AMSC N/A FSC 5975
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TABLE I. Stuffing tube assembly (steel) for bulkheads.

| Assembly part or identifying number | Part or identifying numbers of components | | | Size | Symbol number |
|---|---|---|---|---|---|
| | Tube body 1 required | Gland nut 1 required | Gland ring 1 required | | |
| M24235/18-001 | M24235/18-121 | M24235/9-061 | M24235/9-181 | A | 1821 |
| M24235/18-002 | M24235/18-122 | M24235/9-062 | M24235/9-182 | B | 1822 |
| M24235/18-003 | M24235/18-123 | M24235/9-063 | M24235/9-183 | C | 1823 |
| M24235/18-004 | M24235/18-124 | M24235/9-064 | M24235/9-184 | D | 1824 |
| M24235/18-005 | M24235/18-125 | M24235/9-065 | M24235/9-185 | E | 1825 |
| M24235/18-006 | M24235/18-126 | M24235/9-066 | M24235/9-186 | F | 1826 |
| M24235/18-007 | M24235/18-127 | M24235/9-067 | M24235/9-187 | G | 1827 |
| M24235/18-008 | M24235/18-128 | M24235/9-068 | M24235/9-188 | J | 1828 |
| M24235/18-009 | M24235/18-129 | M24235/9-069 | M24235/9-189 | K | 1829 |
| M24235/18-010 | M24235/18-130 | M24235/9-070 | M24235/9-190 | L | 1830 |
| M24235/18-011 | M24235/18-131 | M24235/9-071 | M24235/9-191 | M | 1831 |
| M24235/18-012 | M24235/18-132 | M24235/9-072 | M24235/9-192 | N | 1832 |
| M24235/18-013 | M24235/18-133 | M24235/9-073 | M24235/9-193 | P | 1833 |
| M24235/18-014 | M24235/18-134 | M24235/9-074 | M24235/9-194 | R | 1834 |
| M24235/18-015 | M24235/18-135 | M24235/9-075 | M24235/9-195 | S | 1835 |
| M24235/18-016 | M24235/18-136 | M24235/9-076 | M24235/9-196 | T | 1836 |
| M24235/18-017 | M24235/18-137 | M24235/9-077 | M24235/9-197 | V | 1837 |
| M24235/18-018 | M24235/18-138 | M24235/9-078 | M24235/9-198 | W | 1838 |
| M24235/18-019 | M24235/18-139 | M24235/9-079 | M24235/9-199 | X | 1839 |
| M24235/18-020 | M24235/18-140 | M24235/9-080 | M24235/9-200 | Y | 1840 |
| M24235/18-021 | M24235/18-141 | M24235/9-081 | M24235/9-201 | Z | 1841 |
| M24235/18-022 | M24235/18-142 | M24235/9-082 | M24235/9-202 | AA | 1842 |
| M24235/18-023 | M24235/18-143 | M24235/9-083 | M24235/9-203 | BB | 1843 |

TABLE II. Stuffing tube assembly (aluminum) for bulkheads.

| Assembly part or identifying number | Part or identifying numbers of components | | | Size | Symbol number |
|---|---|---|---|---|---|
| | Tube body 1 required | Gland nut 1 required | Gland ring 1 required | | |
| M24235/18-031 | M24235/18-151 | M24235/17-241 | M24235/17-271 | A | 1871 |
| M24235/18-032 | M24235/18-152 | M24235/17-242 | M24235/17-272 | B | 1872 |
| M24235/18-033 | M24235/18-153 | M24235/17-243 | M24235/17-273 | C | 1873 |
| M24235/18-034 | M24235/18-154 | M24235/17-244 | M24235/17-274 | D | 1874 |
| M24235/18-035 | M24235/18-155 | M24235/17-245 | M24235/17-275 | E | 1875 |
| M24235/18-036 | M24235/18-156 | M24235/17-246 | M24235/17-276 | F | 1876 |
| M24235/18-037 | M24235/18-157 | M24235/17-247 | M24235/17-277 | G | 1877 |
| M24235/18-038 | M24235/18-158 | M24235/17-248 | M24235/17-278 | J | 1878 |
| M24235/18-039 | M24235/18-159 | M24235/17-249 | M24235/17-279 | K | 1879 |
| M24235/18-040 | M24235/18-160 | M24235/17-250 | M24235/17-280 | L | 1880 |



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TABLE II. Stuffing tube assembly (aluminum) for bulkheads - Continued.

| Assembly part or identifying number | Part or identifying numbers of components | | | Size | Symbol number |
|---|---|---|---|---|---|
| | Tube body 1 required | Gland nut 1 required | Gland ring 1 required | | |
| M24235/18-041 | M24235/18-161 | M24235/17-251 | M24235/17-281 | M | 1881 |
| M24235/18-042 | M24235/18-162 | M24235/17-252 | M24235/17-282 | N | 1882 |
| M24235/18-043 | M24235/18-163 | M24235/17-253 | M24235/17-283 | P | 1883 |
| M24235/18-044 | M24235/18-164 | M24235/17-254 | M24235/17-284 | R | 1884 |
| M24235/18-045 | M24235/18-165 | M24235/17-255 | M24235/17-285 | S | 1885 |
| M24235/18-046 | M24235/18-166 | M24235/17-256 | M24235/17-286 | T | 1886 |
| M24235/18-047 | M24235/18-167 | M24235/17-257 | M24235/17-287 | V | 1887 |
| M24235/18-048 | M24235/18-168 | M24235/17-258 | M24235/17-288 | W | 1888 |
| M24235/18-049 | M24235/18-169 | M24235/17-259 | M24235/17-289 | X | 1889 |
| M24235/18-050 | M24235/18-170 | M24235/17-260 | M24235/17-290 | Y | 1890 |
| M24235/18-051 | M24235/18-171 | M24235/17-261 | M24235/17-291 | Z | 1891 |
| M24235/18-052 | M24235/18-172 | M24235/17-262 | M24235/17-292 | AA | 1892 |
| M24235/18-053 | M24235/18-173 | M24235/17-263 | M24235/17-293 | BB | 1893 |

TABLE III. Stuffing tube assembly (steel) for decks.

| Assembly part or identifying number | Part or identifying numbers of components | | | Size | Symbol number |
|---|---|---|---|---|---|
| | Tube body 1 required | Gland nut 1 required | Gland ring 1 required | | |
| M24235/18-061 | M24235/18-181 | M24235/9-061 | M24235/9-181 | A | 1911 |
| M24235/18-062 | M24235/18-182 | M24235/9-062 | M24235/9-182 | B | 1912 |
| M24235/18-063 | M24235/18-183 | M24235/9-063 | M24235/9-183 | C | 1913 |
| M24235/18-064 | M24235/18-184 | M24235/9-064 | M24235/9-184 | D | 1914 |
| M24235/18-065 | M24235/18-185 | M24235/9-065 | M24235/9-185 | E | 1915 |
| M24235/18-066 | M24235/18-186 | M24235/9-066 | M24235/9-186 | F | 1916 |
| M24235/18-067 | M24235/18-187 | M24235/9-067 | M24235/9-187 | G | 1917 |
| M24235/18-068 | M24235/18-188 | M24235/9-068 | M24235/9-188 | J | 1918 |
| M24235/18-069 | M24235/18-189 | M24235/9-069 | M24235/9-189 | K | 1919 |
| M24235/18-070 | M24235/18-190 | M24235/9-070 | M24235/9-190 | L | 1920 |
| M24235/18-071 | M24235/18-191 | M24235/9-071 | M24235/9-191 | M | 1921 |
| M24235/18-072 | M24235/18-192 | M24235/9-072 | M24235/9-192 | N | 1922 |
| M24235/18-073 | M24235/18-193 | M24235/9-073 | M24235/9-193 | P | 1923 |
| M24235/18-074 | M24235/18-194 | M24235/9-074 | M24235/9-194 | R | 1924 |
| M24235/18-075 | M24235/18-195 | M24235/9-075 | M24235/9-195 | S | 1925 |
| M24235/18-076 | M24235/18-196 | M24235/9-076 | M24235/9-196 | T | 1926 |
| M24235/18-077 | M24235/18-197 | M24235/9-077 | M24235/9-197 | V | 1927 |
| M24235/18-078 | M24235/18-198 | M24235/9-078 | M24235/9-198 | W | 1928 |
| M24235/18-079 | M24235/18-199 | M24235/9-079 | M24235/9-199 | X | 1929 |
| M24235/18-080 | M24235/18-200 | M24235/9-080 | M24235/9-200 | Y | 1930 |



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TABLE III. Stuffing tube assembly (steel) for decks - Continued.

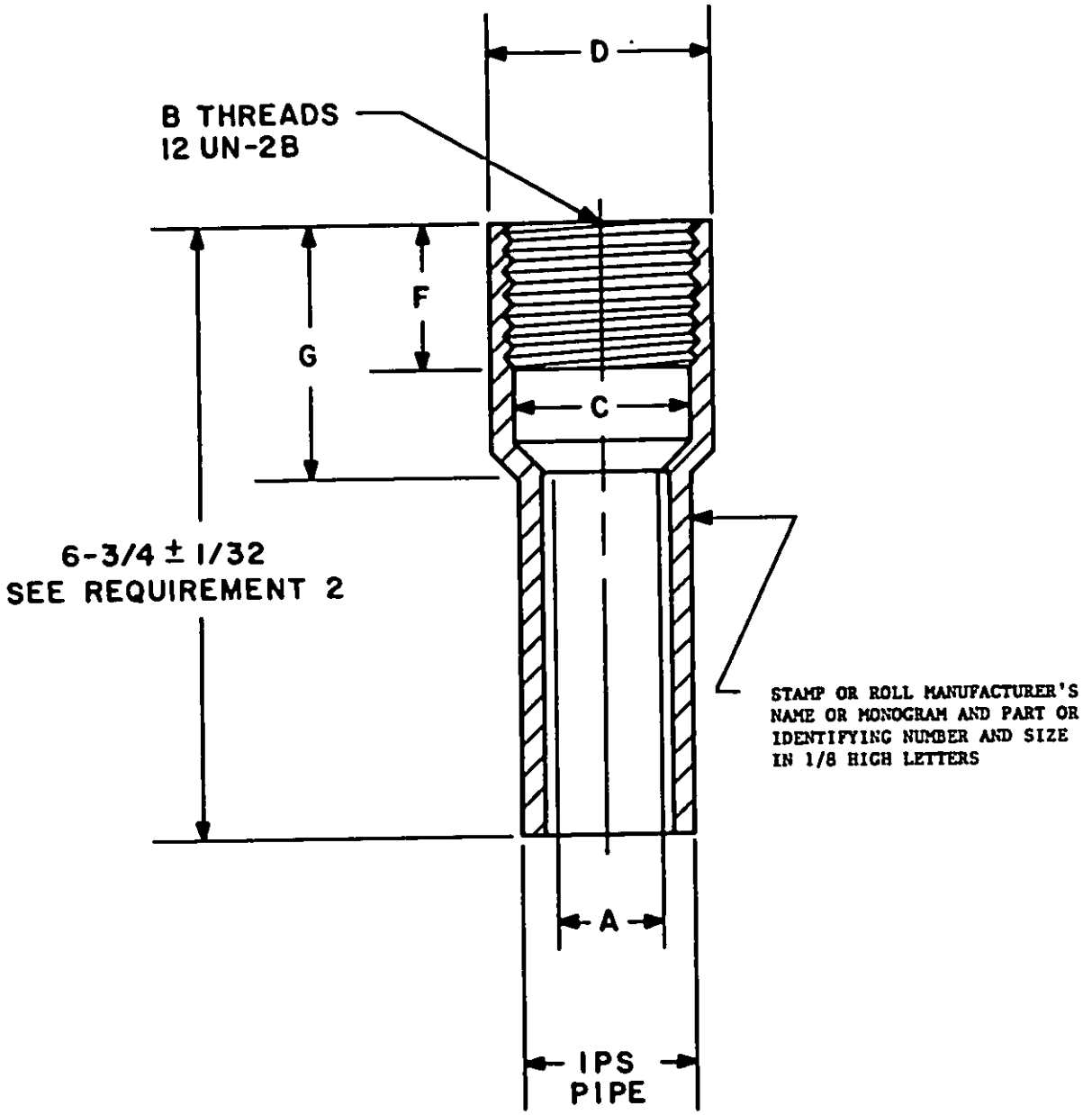
| Assembly part or identifying number | Part or identifying numbers of components | | | Size | Symbol number |
|---|---|---|---|---|---|
| | Tube body 1 required | Gland nut 1 required | Gland ring 1 required | | |
| M24235/18-081 | M24235/18-201 | M24235/9-081 | M24235/9-201 | Z | 1931 |
| M24235/18-082 | M24235/18-202 | M24235/9-082 | M24235/9-202 | AA | 1932 |
| M24235/18-083 | M24235/18-203 | M24235/9-083 | M24235/9-203 | BB | 1933 |

TABLE IV. Stuffing tube assembly (aluminum) for decks.

| Assembly part or identifying number | Part or identifying numbers of components | | | Size | Symbol number |
|---|---|---|---|---|---|
| | Tube body 1 required | Gland nut 1 required | Gland ring 1 required | | |
| M24235/18-091 | M24235/18-211 | M24235/17-241 | M24235/17-271 | A | 1941 |
| M24235/18-092 | M24235/18-212 | M24235/17-242 | M24235/17-272 | B | 1942 |
| M24235/18-093 | M24235/18-213 | M24235/17-243 | M24235/17-273 | C | 1943 |
| M24235/18-094 | M24235/18-214 | M24235/17-244 | M24235/17-274 | D | 1944 |
| M24235/18-095 | M24235/18-215 | M24235/17-245 | M24235/17-275 | E | 1945 |
| M24235/18-096 | M24235/18-216 | M24235/17-246 | M24235/17-276 | F | 1946 |
| M24235/18-097 | M24235/18-217 | M24235/17-247 | M24235/17-277 | G | 1947 |
| M24235/18-098 | M24235/18-218 | M24235/17-248 | M24235/17-278 | J | 1948 |
| M24235/18-099 | M24235/18-219 | M24235/17-249 | M24235/17-279 | K | 1949 |
| M24235/18-100 | M24235/18-220 | M24235/17-250 | M24235/17-280 | L | 1950 |
| M24235/18-101 | M24235/18-221 | M24235/17-251 | M24235/17-281 | M | 1951 |
| M24235/18-102 | M24235/18-222 | M24235/17-252 | M24235/17-282 | N | 1952 |
| M24235/18-103 | M24235/18-223 | M24235/17-253 | M24235/17-283 | P | 1953 |
| M24235/18-104 | M24235/18-224 | M24235/17-254 | M24235/17-284 | R | 1954 |
| M24235/18-105 | M24235/18-225 | M24235/17-255 | M24235/17-285 | S | 1955 |
| M24235/18-106 | M24235/18-226 | M24235/17-256 | M24235/17-286 | T | 1956 |
| M24235/18-107 | M24235/18-227 | M24235/17-257 | M24235/17-287 | V | 1957 |
| M24235/18-108 | M24235/18-228 | M24235/17-258 | M24235/17-288 | W | 1958 |
| M24235/18-109 | M24235/18-229 | M24235/17-259 | M24235/17-289 | X | 1959 |
| M24235/18-110 | M24235/18-230 | M24235/17-260 | M24235/17-290 | Y | 1960 |
| M24235/18-111 | M24235/18-231 | M24235/17-261 | M24235/17-291 | Z | 1961 |
| M24235/18-112 | M24235/18-232 | M24235/17-262 | M24235/17-292 | AA | 1962 |
| M24235/18-113 | M24235/18-233 | M24235/17-263 | M24235/17-293 | BB | 1963 |



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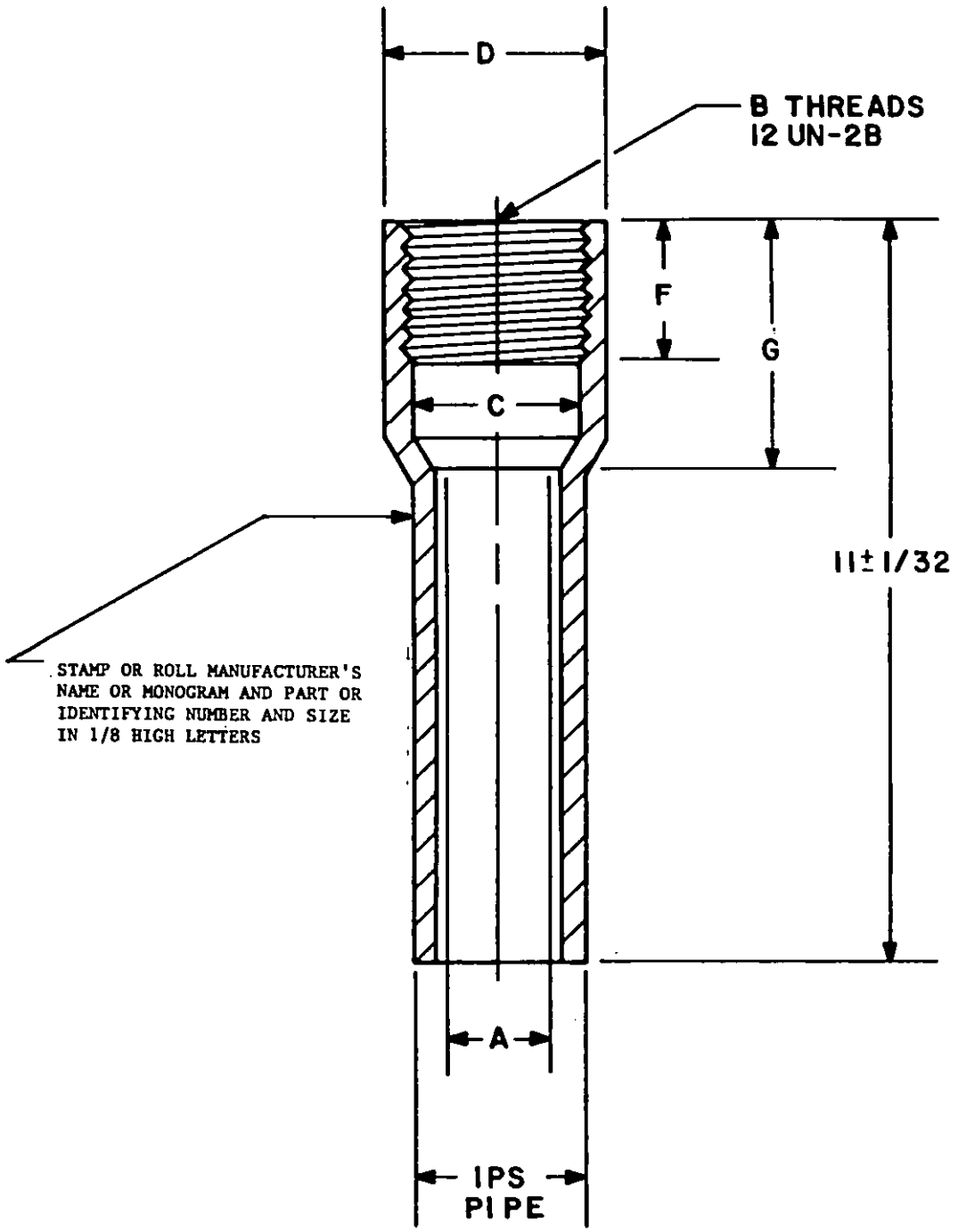



SH 131866

FIGURE 2. Tube body (bulkhead).



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FIGURE 3. Tube body (deck).



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TABLE V. Tube body dimensions.

| Tube body (bulkhead) see figure 2 | | Tube body (deck) see figure 3 | | Size | NPS pipe size | A DIA - 0 +.010 | B THDS | C DIA - 0 +.010 | D DIA "ref" | P 1/ | G |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Part number (steel) | Part number (aluminum) | Part number (steel) | Part number (aluminum) | | | | | | | | |
| M24235/18-121 | M24235/18-151 | M24235/18-181 | M24235/18-211 | A | 1/2 SCH 80-(27/32) | 0.406 | 0.875 | 0.785 | 1.039 | 7/8 | 1-5/8 |
| M24235/18-122 | M24235/18-152 | M24235/18-182 | M24235/18-212 | B | 1/2 SCH 80-(27/32) | .515 | .000 | 1.910 | 1.164 | 7/8 | 1-5/8 |
| M24235/18-123 | M24235/18-153 | M24235/18-183 | M24235/18-213 | C | 3/4 SCH 40-(1-1/16) | .640 | 1.125 | 1.035 | 1.261 | 7/8 | 1-5/8 |
| M24235/18-124 | M24235/18-154 | M24235/18-184 | M24235/18-214 | D | 3/4 SCH 40-(1-1/16) | .750 | 1.250 | 1.160 | 1.386 | 7/8 | 1-5/8 |
| M24235/18-125 | M24235/18-155 | M24235/18-185 | M24235/18-215 | E | 1 SCH 40-(1-21/64) | .812 | 1.250 | 1.160 | 1.426 | 7/8 | 1-5/8 |
| M24235/18-126 | M24235/18-156 | M24235/18-186 | M24235/18-216 | F | 1 SCH 40-(1-21/64) | .843 | 1.312 | 1.222 | 1.488 | 7/8 | 1-5/8 |
| M24235/18-127 | M24235/18-157 | M24235/18-187 | M24235/18-217 | G | 1 SCH 40-(1-21/64) | .953 | 1.500 | 1.410 | 1.676 | 1 | 1-11/16 |
| M24235/18-128 | M24235/18-158 | M24235/18-188 | M24235/18-218 | J | 1-1/4 SCH 40-(1-21/32) | 1.062 | 1.625 | 1.532 | 1.815 | 1 | 1-11/16 |
| M24235/18-129 | M24235/18-159 | M24235/18-189 | M24235/18-219 | K | 1-1/4 SCH 40-(1-21/32) | 1.171 | 1.750 | 1.660 | 1.940 | 1-5/16 | 3 |
| M24235/18-130 | M24235/18-160 | M24235/18-190 | M24235/18-220 | L | 1-1/4 SCH 40-(1-21/32) | 1.265 | 1.812 | 1.720 | 2.000 | 1-5/16 | 3 |
| M24235/18-131 | M24235/18-161 | M24235/18-191 | M24235/18-221 | M | 1-1/2 SCH 40-(1-29/32) | 1.406 | 2.000 | 1.910 | 2.200 | 1-5/16 | 3 |
| M24235/18-132 | M24235/18-162 | M24235/18-192 | M24235/18-222 | N | 1-1/2 SCH 40-(1-29/32) | 1.515 | 2.062 | 2.000 | 2.290 | 1-5/16 | 3 |
| M24235/18-133 | M24235/18-163 | M24235/18-193 | M24235/18-223 | P | 2 SCH 40-(1-5/16) | 1.625 | 2.188 | 2.098 | 2.406 | 1-5/16 | 3 |
| M24235/18-134 | M24235/18-164 | M24235/18-194 | M24235/18-224 | R | 2 SCH 40-(1-5/16) | 1.750 | 2.312 | 2.222 | 2.530 | 1-5/16 | 3 |
| M24235/18-135 | M24235/18-165 | M24235/18-195 | M24235/18-225 | S | 2 SCH 40-(1-5/16) | 1.875 | 2.688 | 2.597 | 2.905 | 1-7/8 | 5 |
| M24235/18-136 | M24235/18-166 | M24235/18-196 | M24235/18-226 | T | 2-1/2 SCH 40-(2-7/8) | 2.062 | 2.875 | 2.785 | 3.191 | 1-7/8 | 5 |
| M24235/18-137 | M24235/18-167 | M24235/18-197 | M24235/18-227 | V | 2-1/2 SCH 40-(2-7/8) | 2.187 | 3.000 | 2.910 | 3.316 | 1-7/8 | 5 |
| M24235/18-138 | M24235/18-168 | M24235/18-198 | M24235/18-228 | W | 2-1/2 SCH 40-(2-7/8) | 2.312 | 3.125 | 3.035 | 3.441 | 1-7/8 | 5 |
| M24235/18-139 | M24235/18-169 | M24235/18-199 | M24235/18-229 | X | 3 SCH 40-(3-1/2) | 2.500 | 3.250 | 3.160 | 3.592 | 1-7/8 | 5 |
| M24235/18-140 | M24235/18-170 | M24235/18-200 | M24235/18-230 | Y | 3 SCH 40-(3-1/2) | 2.609 | 3.375 | 3.285 | 3.717 | 1-7/8 | 5 |
| M24235/18-141 | M24235/18-171 | M24235/18-201 | M24235/18-231 | Z | 3 SCH 40-(3-1/2) | 2.781 | 3.562 | 3.472 | 3.904 | 1-7/8 | 5 |
| M24235/18-142 | M24235/18-172 | M24235/18-202 | M24235/18-232 | AA | 3 SCH 40-(3-1/2) | 2.875 | 3.750 | 3.660 | 4.092 | 2-11/16 | 6-3/4 |
| M24235/18-143 | M24235/18-173 | M24235/18-203 | M24235/18-233 | BB | 3-1/2 SCH 40-(4) | 3.157 | 4.000 | 3.910 | 4.362 | 2-11/16 | 6-3/4 |

1/ Tolerance shall be minus 0, plus 4 threads.





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TABLE V. Tube body dimensions - Continued.

| Tube body (bulkhead) see figure 2 | | Tube body (deck) see figure 3 | | Size | NPS pipe size | A DIA | B THDS | C DIA | D DIA "ref" | F 1/ | G |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Part or identifying number (steel) | Part or identifying number (aluminum) | Part or identifying number (steel) | Part or identifying number (aluminum) | | | | | | | | |
| M24235/18-136 | M24235/18-166 | M24235/18-196 | M24235/18-226 | T | 2-1/2 SCH 40-(2-7/8) | 2.062 | 2.875 | 2.785 | 3.191 | 1-7/8 | 5 |
| M24235/18-137 | M24235/18-167 | M24235/18-197 | M24235/18-227 | V | 2-1/2 SCH 40-(2-7/8) | 2.187 | 3.000 | 2.910 | 3.316 | 1-7/8 | 5 |
| M24235/18-138 | M24235/18-168 | M24235/18-198 | M24235/18-228 | W | 2-1/2 SCH 40-(2-7/8) | 2.312 | 3.125 | 3.035 | 3.441 | 1-7/8 | 5 |
| M24235/18-139 | M24235/18-169 | M24235/18-199 | M24235/18-229 | X | 3 SCH 40-(3-1/2) | 2.500 | 3.250 | 3.160 | 3.592 | 1-7/8 | 5 |
| M24235/18-140 | M24235/18-170 | M24235/18-200 | M24235/18-230 | Y | 3 SCH 40-(3-1/2) | 2.609 | 3.375 | 3.285 | 3.717 | 1-7/8 | 5 |
| M24235/18-141 | M24235/18-171 | M24235/18-201 | M24235/18-231 | Z | 3 SCH 40-(3-1/2) | 2.781 | 3.562 | 3.472 | 3.904 | 1-7/8 | 5 |
| M24235/18-142 | M24235/18-172 | M24235/18-202 | M24235/18-232 | AA | 3 SCH 40-(3-1/2) | 2.875 | 3.750 | 3.660 | 4.092 | 2-11/16 | 6-3/4 |
| M24235/18-143 | M24235/18-173 | M24235/18-203 | M24235/18-233 | BB | 3-1/2 SCH 40-(4) | 3.157 | 4.000 | 3.910 | 4.362 | 2-11/16 | 6-3/4 |

1/ Tolerance shall be minus 0, plus 4 threads.





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REQUIREMENTS:

1. Overall length for bulkhead tube body sizes S thru BB shall be 9 inches in lieu of 6-3/4 inches.
2. Steel bulkhead and deck tubes shall be annealed after swaging.
3. MIL-P-16685, class 1, or class 2 (preformed) shall be used to pack stuffing tubes in service and in tests.
4. Size "BB" stuffing tube is designed to accommodate radio frequency transmission line MIL-L-17342 and Drawing S6702-1338683.
5. Intended use: The stuffing tubes are for electric cable penetrations of surface ships deck and bulkheads where pipe protection is required (see DOD-STD-2003). Alternate to stuffing tubes, MIL-S-24235/9 and MIL-S-24235/17.
6. Wall thickness (after swaging and threading) shall be at least 0.062 inch.

TABLE VI. Parts list.

| Item no. | Description 1/ | Qty 2/ | | Material and specification | Remarks |
|---|---|---|---|---|---|
| | | STL | AL | | |
| 01 | Stuffing tube, swage type | - | -- | -- | See figure 1 and tables I through IV |
| 02 | Tube body (bulkhead) | 1 | -- | Steel, grade A, ASTM A 53 | See figure 2 and table V |
| 03 | Tube body (bulkhead) | - | 1 | Aluminum 5086-H32, WW-T-700/5 or ASTM B 210 | See figure 2 and table V |
| 04 | Tube body (deck) | 1 | -- | Steel, grade A, ASTM A 53 | See figure 3 and table V |
| 05 | Tube body (deck) | - | 1 | Aluminum 5086-H32, WW-T-700/5 or ASTM B 210 | See figure 3 and table V |
| 06 | Gland nut | - | 1 | Aluminum, MIL-S-24235/17 | See tables II and IV |
| 07 | Gland ring | - | 1 | Aluminum, MIL-S-24235/17 | See tables II and IV |
| 08 | Gland nut | 1 | -- | Brass, MIL-S-24235/9 | See tables I and III |
| 09 | Gland ring | 1 | -- | Steel, MIL-S-24235/9 | See tables I and III |

See footnotes at top of next page.



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1/ See applicable figure and table listed in "Remarks" column for part numbers and dimensions.
2/ Quantity is for one complete assembly of each size.

Preparing activity:
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